

Question	Answer	Mark	Mark scheme	Additional guidance
22	$x^2 + y^2 = 80$	P1	for process to find gradient of tangent eg $\frac{10-0}{0--20}$ ($=\frac{1}{2}$) or for $20^2 + 10^2 (= 500)$ or start to method to find angle between tangent and x axis, eg $\tan\theta = \frac{10}{20}$	
		P1	for process to find gradient of normal/radius eg $\frac{-1}{\text{"0.5"}}$ ($=-2$) or for $\sqrt{20^2 + 10^2}$ or $\sqrt{500}$ or 22.36... or 22.4 or completes process to find angle between tangent and x axis. eg $\theta = \tan^{-1}\left(\frac{10}{20}\right)$ ($=26.565\dots$)	
		P1	for equation of tangent eg $y = \text{"0.5"}x + 10$ oe or for equation of radius eg $y = \text{"-2"}x$ oe or for using similar triangles eg $\frac{r}{10} = \frac{20}{\text{"}\sqrt{500}\text{"}}$ or for $\sin(\text{"26.565..."}) = \frac{r}{20}$	
		P1	for process to find the x coordinate eg $\text{"0.5"}x + 10 = \text{"-2"}x$ ($x = -4$) or for $r = \frac{20}{\text{"}\sqrt{500}\text{"}} \times 10$ or $r = 20 \times \sin(\text{"26.565..."})$	
		A1	oe	Accept $(4\sqrt{5})^2$ for 80